

Total flavonoids of *Desmodium styracifolium* attenuates the formation of hydroxy-L-proline-induced calcium oxalate urolithiasis in rats

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Received: 6 November 2016 / Accepted: 24 May 2017 / Published online: 31 May 2017
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Abstract *Desmosium styracifolium* (*D. styracifolium*), which is considered as a Chinese herbal medicine, has been reported to treat the kidney stone diseases. However, the potential phytochemically active components and the underlying mechanisms associated with its efficacy in targeting urolithiasis remain to be elucidated. This study aims to investigate the anti-urolithiatic effect of total flavonoids of *D. styracifolium* (TFDS) on calcium oxalate (CaOx) renal stones in Sprague–Dawley rats. Animal models of CaOx urolithiasis were established in male Sprague–Dawley rats by adding 5% w/w hydroxy-L-proline (HLP) in regular rat chow. The TFDS orally at 100, 400 mg/kg, respectively, were administered along with HLP for 28 days. At the end of 28 days of treatment, urine and serum samples

were collected for crystalluria determination and various biochemical analysis. Kidney tissues were isolated and processed for antioxidant parameters measurement and histopathological examinations. HLP-induced hyperoxaluria alone reliably caused CaOx nephrolithiasis in rats. We showed that TFDS significantly reduced crystalluria and CaOx crystal deposits in the kidney sections as compared to untreated HLP group. Also, TFDS was observed to decrease urinary oxalate excretion, alleviate the proacidosis condition, improve the impaired renal functions and renal epithelial cell injury. Moreover, TFDS protected against the oxidative stress changes via reducing MDA content, increasing CAT and GSH-Px activities in renal homogenate, as well as attenuating the expression of MCP-1, OPN and TGF- β proteins. These results indicated that TFDS had beneficial effect on inhibition of CaOx formation in the rat kidney probably through a combination of antioxidant, anti-inflammatory, urine alkalinizing activities, and lowering the concentration of urinary stone-forming constituents. Thus, TFDS might have clinical implications in preventing oxidative renal cell injury and, ultimately, kidney stone formation. The data provide a rationale for the medicinal use of TFDS in nephrolithiasis and identify this agent as a potential source of new antiurolithic drugs.

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Electronic supplementary material The online version of this article (doi:10.1007/s00240-017-0985-y) contains supplementary material, which is available to authorized users.

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Keywords Total flavonoids of *Desmodium styracifolium* · Hydroxy-L-proline · Calcium oxalate · Antioxidant

Introduction

Urolithiasis, characterized by biomineralization within the urinary tract, is one of the most common urological disorders worldwide with a high rate of prevalence and recurrence [1]. Calcium oxalate (CaOx) has been shown as the

major constituent of urolithiasis, and hyperoxaluria is considered to be one of the predominant risk factors for CaOx kidney stone [2]. Endoscopic stone removal techniques and extracorporeal shock wave lithotripsy have revolutionized the treatment of nephrolithiasis, but they are costly with high rate of recurrence in stone formation, which is around 50% in 5–10 years and up to 75% within 20 years of follow-up [3]. Thus, the development of non-surgical methods to prevent the recurrence of urolithiasis is urgently demanded. With the understanding of the pathophysiological characteristics and the mechanisms of herbal remedies towards lithogenesis, phytotherapy is rolling out as an alternative treatment of kidney stone [4, 5]. Traditional Chinese medicine (TCM) plays an important role in preventing the formation of urolithiasis through against increasing excretion of urinary citrate, decreasing excretion of urinary calcium and oxalate, and diuretic effect [6]. Nevertheless, the detailed molecular mechanisms by which TCM in treating the kidney stone remains poorly understood.

Desmosium styracifolium (Osb.) Merr. (Guang-Jin-Qian-Cao in Chinese), a plant member of genus *Desmodium* (Fabaceae), is widely used in TCM for the treatment of kidney stone diseases because of its heat-clearing as well as diuretic properties [7]. Ethnobotanical literature since ancient China revealed that *D. styracifolium* has been used in folk medicine for the treatment of urolithiasis, cardiovascular diseases and hepatitis. It is an important medicinal herb in treating kidney stones, which has been attributed to polysaccharides, triterpenoids, and flavonoids [7]. Phytochemical studies have indicated that *D. styracifolium* contains flavonoid compounds such as viciensins and isovitexin [8]. In vitro crystallization study has demonstrated the prophylactic effect of *D. styracifolium* in CaOx crystals formation [9], and further in vivo study subsequently confirmed that the aqueous extracts of *D. styracifolium* inhibited the formation of CaOx stone via increasing urinary citrate excretion, reducing urinary calcium, diuresis, as well as antioxidative effects in rats [10]. We have also previously demonstrated the antiurolithic and antioxidant properties of aqueous crude extracts and different polarity fractions of *D. styracifolium* in vivo, which were consistent with the above [11–13]. However, the potential phytochemically active components and the underlying mechanisms associated with its efficacy in targeting urolithiasis are largely unknown.

Reactive oxygen species (ROS), mainly composed of free radicals, molecules or atoms contained unpaired electrons and their metabolites, has been to be served as mediators in lots of regulatory processes and biological events, including the growth, activation or inactivation of biomolecules as well as regulation of transcriptional activity [14]. Accumulating data have shown that renal cellular exposure to oxalate and/or CaOx crystals leads to generation of ROS,

triggering oxidative stress together with renal epithelial cell injury, which directly aggravates CaOx crystal formation and deposition in the renal tubules [15, 16]. Several macromolecules have been detected to modulate the process including angiotensin II (Ang II), transforming growth factor- β (TGF- β), monocyte chemoattractant protein-1 (MCP-1) and osteopontin (OPN) [17, 18]. Also, a series of tissue culture and animal model studies have demonstrated that treatments with anti-oxidants and free radical scavengers reduce oxalate and CaOx crystal induced injuries and CaOx crystal deposition in kidneys [14]. These results indicate a great potential for the therapeutic application of antioxidants and free radical scavengers to reduce crystal formation and stone recurrence.

In this study, we investigated the antiurolithiatic effects of total flavonoids of *Desmodium styracifolium* (TFDS) on CaOx renal stones induced by administration of hydroxy-L-proline (HLP) in rats.

Materials and methods

Chemicals and reagents

HLP (PubChem CID: 5810) was purchased from Weijia Chemical Co. Ltd. (Guangdong, China). Oxalate and citrate were obtained from Kemiou Chemical Co. Ltd. (Tianjin, China). Kits used in this study for the determination of calcium, magnesium, creatinine and blood urea nitrogen (BUN) were supplied by Roche Co. Ltd. (Shanghai, China). Commercially available kits of malondialdehyde (MDA), superoxide dismutase (SOD), catalase (CAT) and glutathione peroxidase (GSH-Px) were purchased from Nanjing Jiancheng bioengineering Institute (Jiangsu, China). Reagents used for histological preparations were hematoxylin and eosin (H&E) from Fuzhou Maixin Biotech. Co. Ltd. (Fujian, China). The polyclonal antibody against OPN, TGF- β and MCP-1 was obtained from Boster Bioengineering Institute (Hubei, China). All the chemicals used were of analytical grade available.

Plant materials

The aerial parts of *D. styracifolium* (Batch No. 121106151) were procured from Kangmei Pharmaceutical CO. LTD. (Guangdong, China), and were authenticated by one of the authors (Zhao Z). A voucher specimen (No. 2013034) has been deposited at the herbarium of School of Chinese Herbal Medicine, Guangzhou University of Chinese Medicine. The extraction and purification of TFDS were carried out according to our previous report [19]. Briefly, the finely powdered plant material was refluxed twice with 50% ethanol for 1.5 h each at 80 °C. After filtration through the

filter paper, the extract solution was combined and diluted (total flavonoids content 8.0 mg/mL), and the pH value was adjusted to 5. Then the solution was absorbed to DM-130 macroporous resin and eluted with 50% ethanol. The eluent was pooled, concentrated, and spray-dried to obtain TFDS.

Animal models

A total of 40 male Sprague–Dawley rats, weighing approximately 180–220 g, were purchased from Guangdong Medical Laboratory Animal Center (Foshan, Guangdong). After 1 week of acclimation under the controlled condition of room temperature at 25 °C with a 12-h light/dark cycle, the rats were then randomly divided into five groups of eight each. Animal model of CaOx kidney stone was established by administration of food which was supplemented with HLP to 5% (w/w) [20, 21]. The experimental groups are summarized below: Normal control group, rats were fed a normal rat chow and de-ionized water; TFDS control group, rats received TFDS (400 mg/kg) by intragastric administration except for normal rat chow and water; Model control group, rats received a specialized chow supplemented with HLP to 5% (w/w) and de-ionized water; TFDS 100 treated group, rats received a diet similar to model control group rats except orally administration of TFDS (100 mg/kg); TFDS 400 treated group, rats received a diet similar to model control group rats except orally administration of TFDS (400 mg/kg). All animals had free access to drinking water and chow. The condition of the animals, consumption of food and water intake were recorded every day, and the weight of rats was conducted weekly to check the growth. The experimental procedures were approved by the institutional Animal Ethics Committee of Guangdong Provincial Hospital of Chinese Medicine, in accordance with National Institutes of Health and Nutrition Guidelines for the Care and Use of Laboratory Animals.

Samples collection

No animals died or showed severe clinical symptoms before reaching the 28 days of treatment. At the end of 28 days, the rats were placed into a metabolic cage individually to collect urine samples. The 3-h morning urine was firstly collected for crystalluria study using a commercial counting plate, followed by the 24-h urine for diverse analysis. After determination of urine volume and pH, the urine samples were acidified to pH = 2 with 5 M HCl and centrifuged at 1500 rpm for 10 min to remove debris and the supernatants were stored at −20 °C until analysis. The rats were anesthetized intraperitoneally with 10% chloral hydrate, and then the blood was collected through inferior vena cava. After centrifugation at 4 °C, the serum separation was transferred into clean tubes and stored at −20 °C

for oxidative stress analysis. Both the kidneys were excised and rinsed in ice cold physiological saline. The right kidneys were fixed in 10% neutral buffered formalin for histopathological analysis. The left kidneys were stored at −20 °C for oxidative stress analysis.

Biochemical analysis

Urinary calcium, magnesium, blood urea nitrogen (BUN) and creatinine contents were determined by using commercially kits with automatic biochemical analyzer. For detection of urinary oxalate and citrate, the acidified urine were performed by ion chromatography method [22] using a Dionex ICS-2500 gradient system equipped with a 0.4 × 25 cm IonPac AS11-HC anion analytical column and a 0.4 × 5 cm IonPac AG11-HC guard column. MDA, SOD, CAT and GSH-Px in the left kidney samples were examined spectrophotometrically according to the manufacturer's protocols.

Histopathological examination

The right kidney samples were fixed in 10% formalin, embedded in paraffin, and cut into 5-μm sections, which were processed for hematoxylin and eosin (H&E) stained analysis and immunohistochemistry examination. CaOx crystal deposits were counted in H&E stained slides in 100 × microscope fields under polarized light, using a semiquantitative scoring method according to the previous studies [23]. Briefly, a sagittal section of each renal specimen was divided into eight equal-sized regions by four virtual lines, severity grades were assigned as 0 ≤ 1 crystal, 1 ≤ 10 crystals, 2 ≤ 30 crystals, 3 ≤ 50 crystals, 4 ≤ 75 crystals and 5 ≥ 75 crystals. The total number of CaOx deposits in each specimen was reported as the average of the eight readings.

Immunohistochemistry analysis

Protein expressions of OPN, TGF-β and MCP-1 were detected by immunohistochemistry as previously described [21]. Briefly, paraffin embedded slides were deparaffinized by xylene immersion followed by dehydration in ethanol. Slides were incubated overnight at 4 °C with primary antibodies against OPN, TGF-β and MCP-1. Isotype controls were performed using mouse IgG. Images were taken using a 200 × light microscope (Leica Microimaging, Germany). The positive cells showed brown articles or clumps in the cytoplasm, and the optical density values were measured for semiquantitative analysis.

Statistical analysis

The data were expressed as mean $\pm$ standard deviation (SD) and were analyzed with one-way ANOVA followed by Newman-Keuls post hoc test for multiple comparisons (GraphPad Prism 5.0, San Diego, CA, USA). The crystal deposit scores were analyzed by the non-parametric Kruskal–Wallis test for inter-group comparisons and by the Wilcoxon rank sum test (Mann–Whitney U) for pair-wise comparisons. The p values of less than 0.05 were considered statistically significant.

Results

Quality evaluation of total flavonoids of *D. styracifolium*

Using rutin as standard reference, the content of total flavonoids in the extract was determined to be 68.3% by the colorimetric method using UV spectrophotometer, as described by Chinese Pharmacopeia. To determine whether the main constituents in the extract were flavonoids, the

extract of TFDS was further analyzed by UHPLC–MS/MS analysis (see Supplemental material part). Combining their characteristic UV absorption, accurate mass measurements within 3 ppm error for each molecular ion and subsequent fragment ions, as well as those data provided in literature [24], the major absorption peaks appeared at 4.2, 4.9, 5.2, 5.7, 6.0 and 7.7 min were identified to be vicenin 2 (1), vicenin 1(2), schaftoside (3), isoschaftoside (4), vicenin 3 (5), isovitexin (6), respectively, which further confirmed that the major constituents of the studied sample were flavonoids (Fig. 1; Table S1).

Effects of TFDS on general and biochemical variables in rats

In the animal study, all the parameters recorded between the TFDS control and normal control groups were not recognizably different. The detailed parameters recorded from groups of animals at the end of 28 days of experimental period are shown in Table 1.

The gain in body weight and food intake were significantly ($p < 0.01$) reduced in stone forming group as

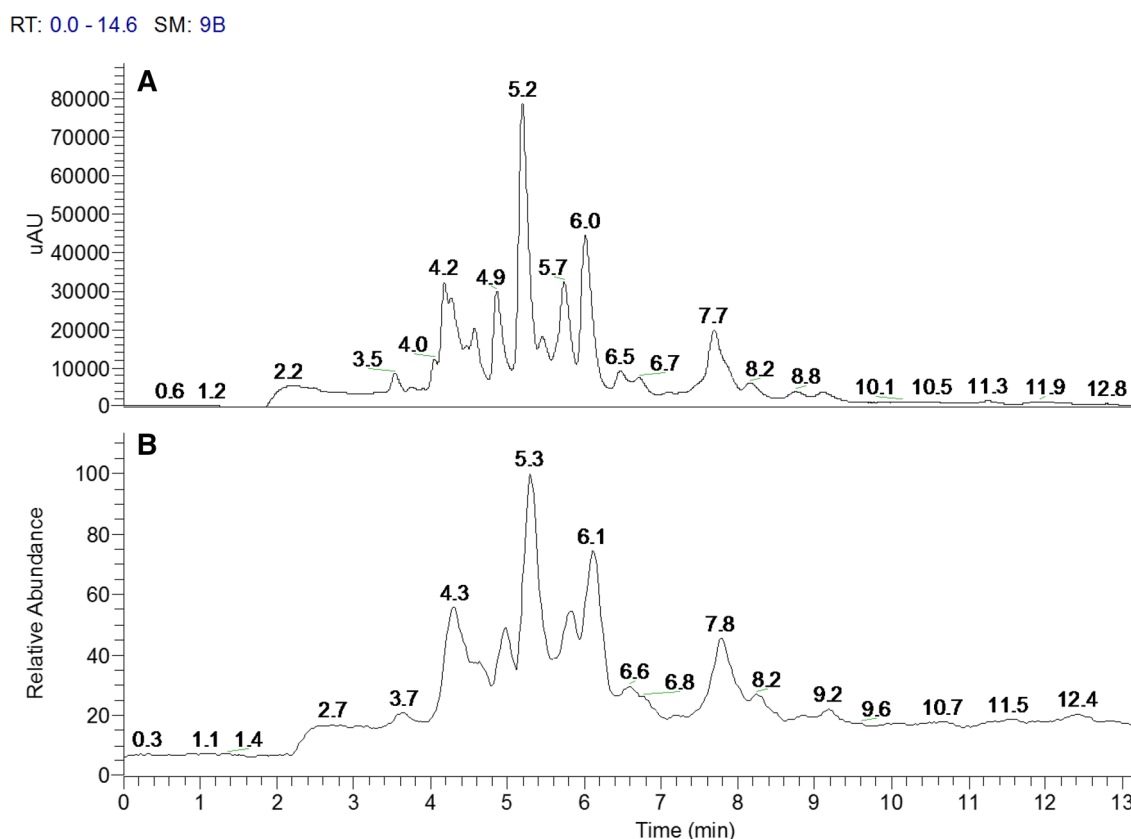


Fig. 1 UHPLC-PAD/ESI-MS analysis of total flavonoids of *D. styracifolium*. **a** UHPLC-PAD chromatography (upper panel); **b** ESI-MS total ion current (lower panel). The major absorption peaks appeared at 4.2, 4.9, 5.2, 5.7, 6.0 and 7.7 min were identified to be

vicenin 2, vicenin 1, schaftoside, isoschaftoside, vicenin 3, isovitexin, respectively, which further confirmed that the major constituents of the studied sample were flavonoids

Table 1 Parameters recorded from urin and blood of groups after 4 weeks of study

Parameters	Normal ctrl	TFDS ctrl	HLP	HLP + TFDS 100 (mg/kg)	HLP + TFDS 400 (mg/kg)
Gain in BW (%)	33.25 ± 8.76	35.83 ± 4.03	4.29 ± 11.88**	7.62 ± 5.29**	13.90 ± 4.04** [#]
Kidney wts (g)	1.20 ± 0.12	1.07 ± 0.93	1.48 ± 0.24**	1.41 ± 0.21*	1.22 ± 0.12 [#]
Water intake (ml)	36.67 ± 7.53	36.67 ± 5.16	48.33 ± 8.76*	35.83 ± 13.20 [#]	37.50 ± 11.29
Food intake (g)	25.50 ± 3.39	25.67 ± 5.89	14.67 ± 2.50**	21.00 ± 5.73 [#]	16.67 ± 3.27**
Urinary/24 h					
Volume (ml)	12.83 ± 2.93	12.50 ± 4.32	26.17 ± 6.52**	26.17 ± 8.95**	18.50 ± 3.42
pH	7.70 ± 0.72	8.17 ± 0.39	6.59 ± 0.47**	7.09 ± 0.36*	7.45 ± 0.33 [#]
CU (count/mm ³)	0	0	7.50 ± 3.78**	2.00 ± 4.00 [#]	0.83 ± 1.33 [#]
Mg ²⁺ (mg)	9.40 ± 2.71	11.80 ± 2.57	5.08 ± 2.60**	7.91 ± 2.01 [#]	5.12 ± 0.91**
Ca ²⁺ (mg)	2.37 ± 1.74	3.23 ± 1.46	0.82 ± 0.19*	2.70 ± 1.14 [#]	2.04 ± 1.02
Oxalate (mg)	7.18 ± 1.73	5.68 ± 1.86	12.62 ± 5.25*	9.44 ± 3.44	6.49 ± 3.36 [#]
Citrate (mg)	83.20 ± 21.39	100.76 ± 29.21	51.01 ± 12.57*	82.57 ± 18.55 [#]	47.82 ± 12.66*
Serum					
Cr (μmol/L)	26.80 ± 4.60	28.60 ± 2.97	80.75 ± 54.86**	45.80 ± 11.71 [#]	45.25 ± 5.32 [#]
BUN (mmol/L)	7.42 ± 0.64	7.32 ± 1.24	16.02 ± 10.80*	11.92 ± .93	9.20 ± 1.65 [#]
AST (U/L)	56.67 ± 22.00	54.00 ± 7.21	40.67 ± 3.98	38.83 ± 5.46*	38.17 ± 4.17*
ALT (U/L)	181.67 ± 21.07	137.67 ± 16.12	132.50 ± 27.16*	104.17 ± 10.09*	94.83 ± 11.05*

The values shown are mean ± SD ($n = 8$)

Statistical analysis was done with one-way ANOVA followed by Newman–Keuls post hoc test for multiple comparisons

HLP hydroxy-L-proline, TFDS total flavonoids of *Desmodium styracifolium*, BW body weight, CU crystalluria, Cr creatinine, BUN blood urea nitrogen

Statistical significance: * $p < 0.05$ versus normal ctrl, ** $p < 0.01$ versus normal ctrl, [#] $p < 0.05$ versus HLP ctrl, ^{##} $p < 0.01$ versus HLP ctrl

compared to the normal control group. The co-administration of TFDS (400 mg/kg) prevented ($p < 0.05$) the loss in body weight. Conversely, kidneys excised from stone forming group were enlarged and heavier ($p < 0.01$) as compared to the normal control group, whereas, TFDS (400 mg/ml) significantly ($p < 0.05$) reduced kidney weights as compared to the stone forming group. The water intake was higher ($p < 0.05$) in the stone forming group compared to that in the normal control animals; however, the food intake was lower ($p < 0.01$) in the stone forming group compared to that in the normal control animals. The 24-h urinary volume was high in the lithogenic group ($p < 0.01$) as compared to the normal control group. TFDS at 400 mg/kg treatment reduced urinary volume as compared to that of lithogenic group, though not to a significant extent. Lithogenic treatment significantly ($p < 0.01$) reduced pH of urine as compared to that of the normal control group, while co-administration of TFDS increases pH in a dose-dependent manner ($p < 0.01$).

As to the urine composition, abundant envelope-like hexahedral CaOx crystals were observed in the 3-h morning urine collected from lithogenic group. TFDS treatment significantly ($p < 0.01$) reduced crystalluria in a dose-dependent manner. In parallel with crystalluria, there was an increased urinary oxalate concentration ($p < 0.01$) from the lithogenic animals. TFDS prevented the changes

in urinary oxalate ($p < 0.01$) in a dose-dependent manner, although the parameter remained higher than those of the control animals at 100 mg/kg. Mg²⁺, Ca²⁺, and the citrate content significantly reduced in the lithogenic animals ($p < 0.01$, $p < 0.05$, $p < 0.05$, respectively), as compared to that of normal control group. TFDS at 100 mg/kg treatment prevented the changes ($p < 0.05$), ignoring the modest effects even at 400 mg/kg.

Lithogenic treatment caused impairment of renal functions of the untreated rats as evident from raising BUN and serum creatinine levels, markers of glomerular and tubular damage. Co-administration of TFDS protects against the damage by reducing BUN and serum creatinine concentrations ($p < 0.05$). On the other hand, the protective effect of TFDS on the renal epithelial cell injury was also seen from the histopathological morphological slides, which was consistent with the blood functional test. In addition, we showed that there was no obvious side effect of TFDS from the view of hepatic function parameters.

TFDS reduced HLP-induced CaOx deposition in the kidneys

Histopathological analysis revealed no CaOx crystalline deposits or other abnormalities in the normal control group and TFDS control group (Fig. 2A1, A2).

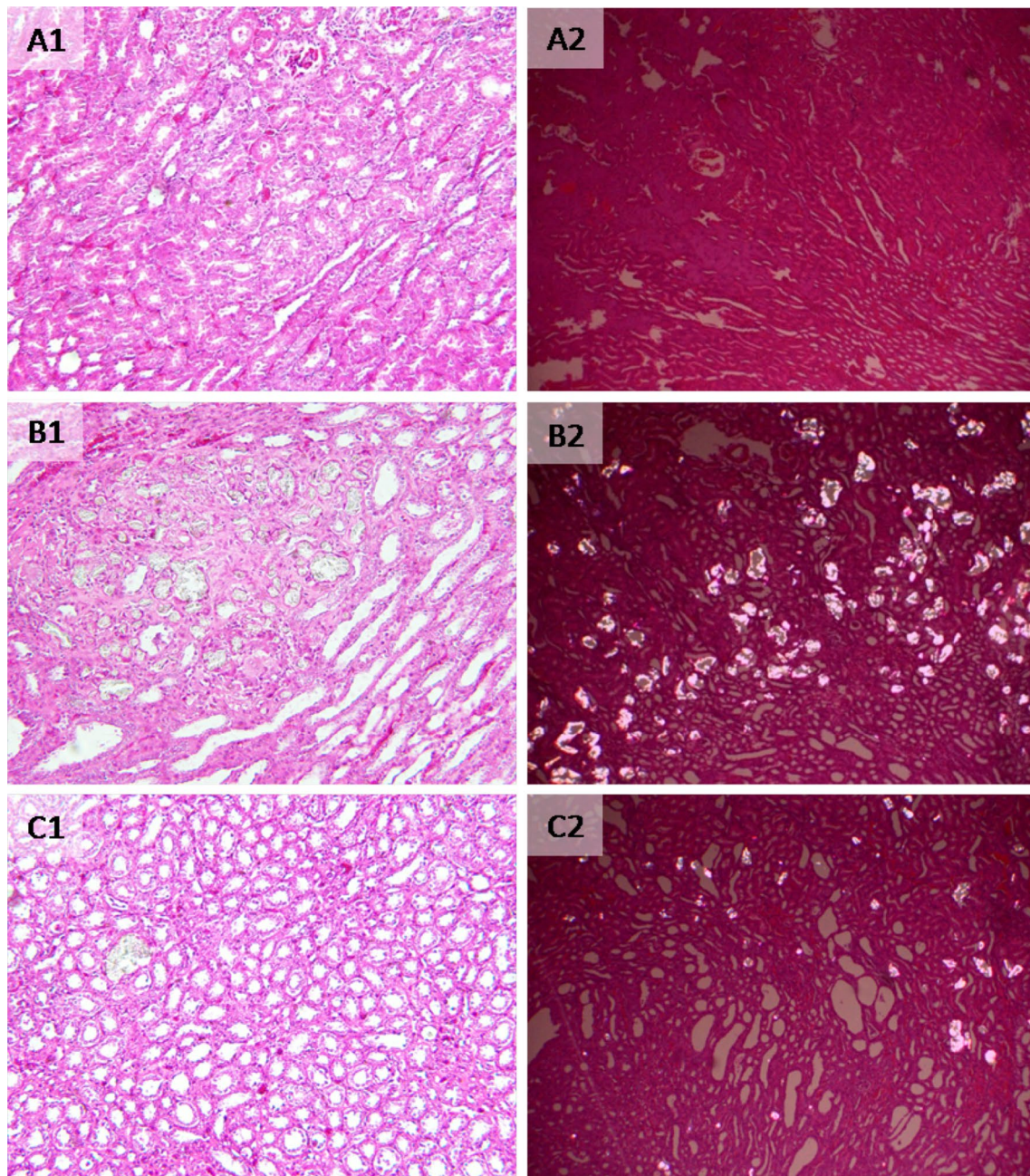


Fig. 2 Representative microscopic images of kidney sections from the animals in study for the preventive effect of TFDS ($\times 100$). **A1**, **B1** and **C1** are H&E stained kidney sections from vehicle control,

stone forming and treated groups (400 mg/kg), respectively and **A2**, **B2** and **C2** are the respective sections under view of the polarized light. The *white spots* point to crystal deposits in the kidney

Whereas, glomerular, peritubular and blood vessel congestion, epithelial cell desquamation, and deposition of inflammatory cells were observed in stone-inducing group (Fig. 2B1). Moreover, lots of crystal deposits were observed under polarized light in all regions of kidneys in lithogenic group (Fig. 2B2). However, co-administration of TFDS not only gradually prevents the renal tissue injuries from lithogenic treatment (Fig. 2C1), but also significantly ($p < 0.01$) reduces

CaOx crystal deposits as compared to untreated HLP group (Figs. 2C2, 3).

TFDS alleviated the effect of oxidative stress response

We then tested the potential mechanisms by which TFDS reduced the kidney stone deposit. ROS have been shown to be involved in many significant regulatory renal cellular events via oxidative stress response, including lipid

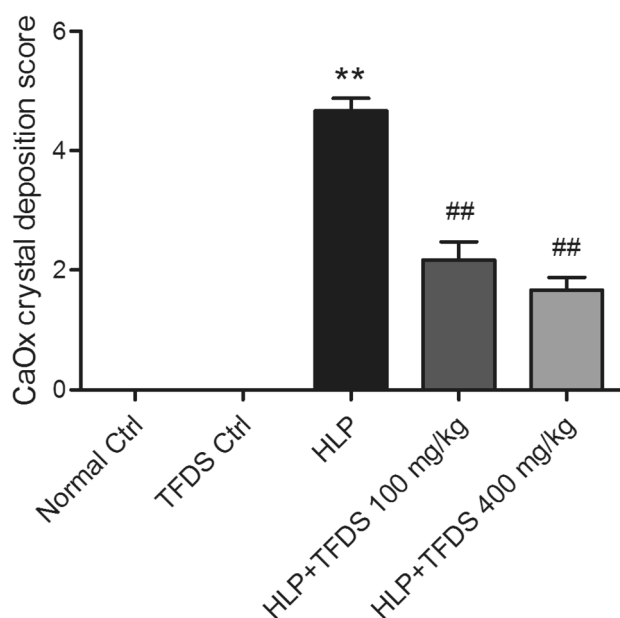


Fig. 3 CaOx crystal deposition scores among groups at the end of the study. The data shown are mean $\pm$ SD ($n = 8$). Severity grade were assigned as $0 \leq 1$ crystal, $1 \leq 10$ crystals, $2 \leq 30$ crystals, $3 \leq 50$ crystals, $4 \leq 75$ crystals and $5 \geq 75$ crystals. Statistical significance ** $p < 0.01$ versus normal ctrl, ## $p < 0.01$ versus HLP ctrl

peroxidation, triggering of renal epithelial cell injury, promoting crystal nucleation, aggregation and retention [15]. Thus, it is a great potential for the therapeutic application of antioxidants and free radical scavengers to reduce stone formation. We showed that stone-inducing treatment enhanced the content of MDA ($p < 0.01$), which is a lipid peroxidation product, decreased activities of the antioxidant enzymes including CAT ($p < 0.01$), and GSH-Px ($p < 0.05$) in kidneys of the untreated rats as compare to kidneys of the control animals. While a simultaneous treatment with TFDS showed protective effect against the oxidative changes induced by lithogenic treatment, interestingly, no significant changes were observed about the activity of SOD (Fig. 4).

TFDS attenuated MCP-1, OPN, and TGF- β proteins expression

Previous studies reported that renal cellular exposure to oxalate and/or CaOx crystals leads to the development of oxidative stress response, followed by renal epithelial cell injury and inflammation, which is characterized by increased expression of specific encoding genes, transcriptional activators and proinflammatory molecules, such as MCP-1, OPN and TGF- β [14, 21, 25]. This was the reason we test the role of MCP-1, OPN and TGF- β . We showed that while the proteins expression was weak in the

renal tissues of the control groups, stone-inducing treatment enhanced ($p < 0.01$) the expression of OPN, MCP-1 and TGF- β proteins in the kidney sections; however, co-administration treatment of TFDS significantly reduced the staining of these proteins ($p < 0.01$) (Fig. 5). Together, the results suggested that inhibition of CaOx formation might be involved in regulation of OPN, MCP-1 and TGF- β expression by TFDS.

Discussion

Due to the complexity of lithogenesis and high recurrence rate of kidney stone, growing attention has been paid to herbal plants to look for effective antiurolithic agents over the years [4, 5]. *D. styracifolium* has been found to inhibit the formation of urolithiasis, decrease urinary oxalate excretion, facilitate output of urine and protect against oxidative stress response [10–12]. Nevertheless, the active constituents and underlying mechanisms associated with its efficacy in treating urolithiasis remain to be elucidated. In this study, we demonstrated that total flavonoids of *D. styracifolium* reduced experimentally induced CaOx deposit in rats through multiple pathways, including lowering urinary stone-forming composition as well as alleviating renal tubular epithelial cell oxidative and inflammatory injury.

Hyperoxaluria has been considered to be a major risk factor in the formation of CaOx kidney stone, as it leads to oxidative stress and renal epithelial cell injury thereby providing a lesion for crystals attachment and ultimately causes crystal deposition in the kidney [14]. In the present study, we established the experimental hyperoxaluria rat model by administration of food supplemented with HLP to 5% (w/w) [20, 21]. In the most common model, ethylene glycol, a precursor of oxalate, is given to rats in their drinking water with or without additional ammonium chloride or vitamin D [23]. However, it is criticized because ethylene glycol, as well as some of its metabolites, is nephrotoxic, which may cause metabolic acidosis within renal tubular cells. Both oxalate and CaOx crystals are also injurious to renal epithelial cells. So, it is difficult to distinguish the effects between ethylene glycol and its metabolites caused by oxalate and CaOx crystals [20]. In this study, we fed the rats with administration of HLP in the chow to develop hyperoxaluria. As anticipated, rats fed with a HLP-free diet remained normoxaluric and devoid of any crystals, while the HLP-supplemented diet led to hyperoxaluria and variable levels of CaOx crystal depositions in all rats, which is consistent with previous studies [21].

The CaOx kidney stones formation is a result of crystal nucleation, growth, aggregation as well as retention in the renal tubular lumens, whose basis is associated with multifactorial processes mainly about urinary supersaturation

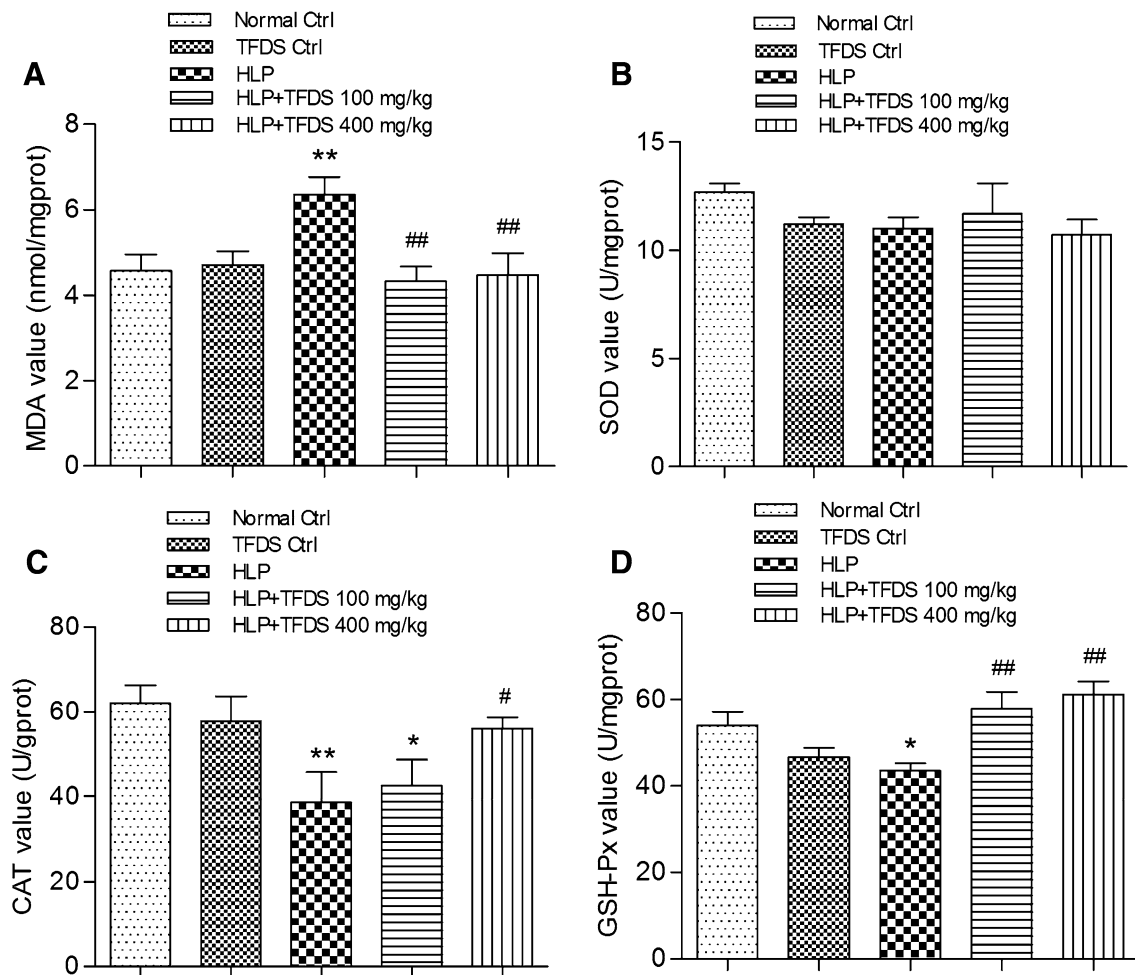


Fig. 4 MDA content and antioxidant enzymes activities in the kidney homogenates among groups. **a** MDA content; **b**, **c** and **d**, SOD, CAT and GSH-Px activities, respectively. The values shown are

mean $\pm$ SD ($n = 8$). Statistical significance * $p < 0.05$ versus normal ctrl, ** $p < 0.01$ versus normal ctrl; # $p < 0.05$ versus HLP ctrl, ## $p < 0.01$ versus HLP ctrl

with stone-forming calcium salts. A number of environmental and metabolic abnormalities such as hyperoxaluria, hypercalciuria, and hypocitraturia, can change the urinary composition and enhance supersaturation of stone-forming salts [26]. In the present study, administration of HLP to rats significantly increased excretion of oxalate, and decreased excretion of citrate, Ca^{2+} and Mg^{2+} in 24-h urine samples of all lithogenic rats. However, oral co-administration of the TFDS significantly decreased urinary oxalate excretion as well as increased contents of citrate, Ca^{2+} and Mg^{2+} . Note that hypercalciuria has been shown to be involved in the formation of CaOx kidney stone in other studies [27], which was different from our current findings. On the contrary, previous studies reported that lithogenic induction with ethylene glycol decreased the urinary Ca^{2+} excretion, whereas medical plants treatment significantly increased the Ca^{2+} levels, which was regarded as one of the underlying mechanisms to treat the urolithiasis [28].

Consistent with this, our current findings demonstrated that lithogenic intervention reduced the urinary Ca^{2+} , but TFDS treatment significantly increased its excretion. Our obtained data showed that urine pH significantly decreased in urolithiasis control group; however, the simultaneous administration of TFDS increased the pH level in a dose-dependent manner. This result suggested that TFDS possess urine-alkalinizing activities, which was similar with previous studies [29]. In addition, the TFDS also exhibited a protective effect against impairment of renal function as indicated by serum creatinine and BUN levels.

Our results demonstrated, for the first time that, oxidative stress in the kidney mediated the effect of TFDS on inhibition of CaOx stone formation. Renal epithelial cell injury by ROS is a prerequisite step in the pathogenesis of urolithiasis [15]. Growing evidence demonstrates that exposure to high levels of oxalate and CaOx crystals promote renal tubular injury through intracellular oxygen

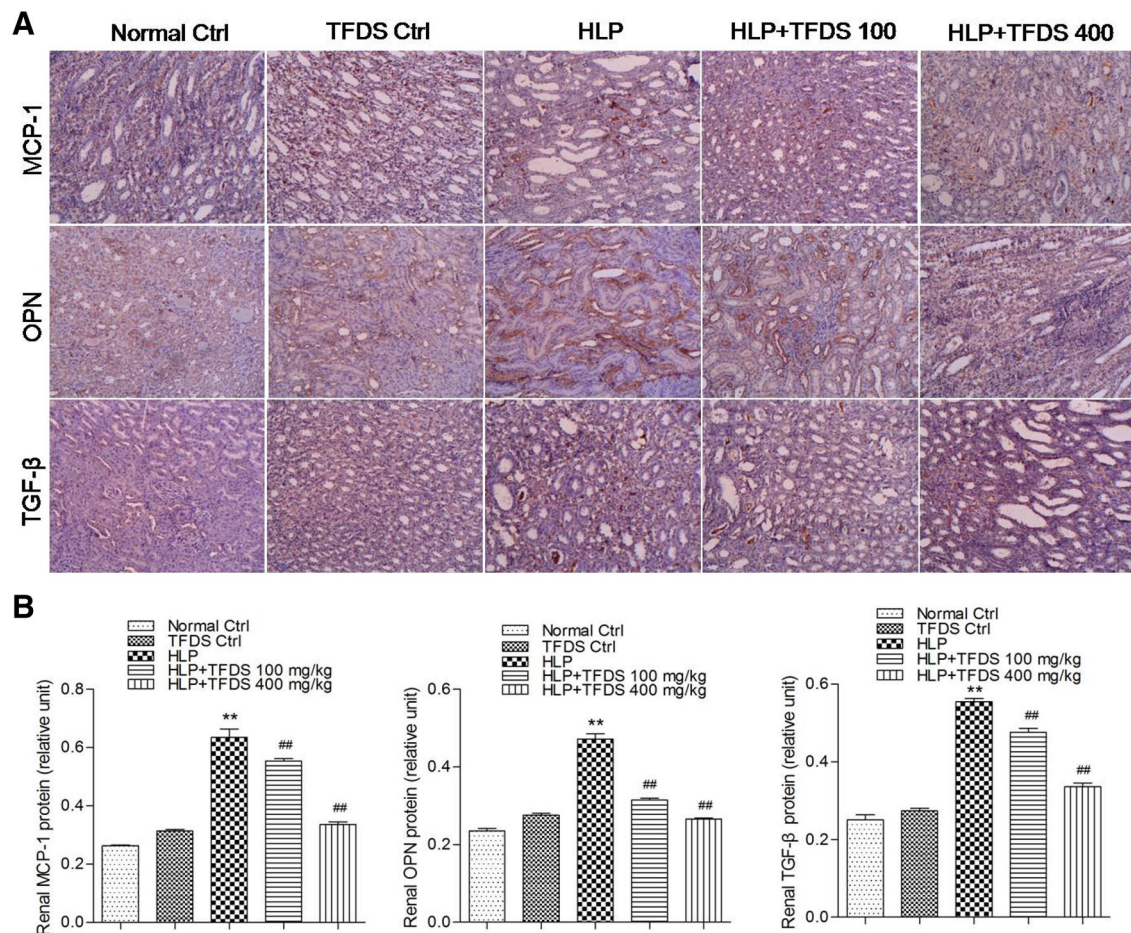


Fig. 5 The expression of MCP-1, OPN and TGF-β in kidney samples from all groups were detected by immunohistochemistry. Representative microscopic images (a) and expression levels (b) of MCP-

1, OPN and TGF-β were shown, respectively. The values shown are mean ± SD ($n = 8$). Statistical significance ** $p < 0.01$ versus normal ctrl; ## $p < 0.01$ versus HLP ctrl

free radical generation, which subsequently facilitates the events of CaOx crystal nucleation, aggregation, retention and stone development [16]. Meanwhile, increased production of ROS triggers the kidney to adapt to this hyperoxaluria insult by up-regulating the antioxidant defense systems, including SOD, CAT, and GSH-Px. Lipid peroxidation has been regarded as the molecular mechanism by which oxalate and/or CaOx promote renal injury in hyperoxaluric rats, characterized by increased levels of MDA [15]. It was shown that flavonoids possess powerful antioxidant properties which decreased the lipid peroxidation production induced by oxalate in renal epithelial cells, consequently preventing the development of renal crystal deposition in rats [30]. Our current findings suggested that TFDS protected against the oxidative changes induced by HLP treatment via reducing MDA content as well as increasing the antioxidant enzymes activities, which was consistent with our previous studies of *D. styracifolium* crude extracts [12, 13]. Nevertheless, further studies are needed to elucidate

the detailed mechanism by which regulation of oxidative stress with TFDS controls CaOx inhibition.

Renal epithelial cells respond to high oxalate and/or CaOx crystals characteristically by up regulation of potential target gene products including transcriptional activators, regulators of extracellular matrix, growth factors as well as inflammatory associated molecules. Intriguingly, our results also suggested the role of MCP-1, OPN and TGF-β in mediating the effect of TFDS on inhibition of CaOx formation. NADPH oxidase is a well-known source of ROS; TGF-β increases the activity of NADPH oxidase by activating Nox2 and Nox4, the homologs of the NADPH oxidase subunit, followed by up regulation of MCP-1 and OPN production [25]. Previous data showed that exposure of renal epithelial cells in culture to high Ox and CaOx crystals leads to increasing expression of MCP-1, OPN and TGF-β [17, 18], which modulates a series of molecular events including induction of localized inflammation by recruitment of monocytes and macrophages, stimulating

extracellular matrix synthesis and promoting adhesion of crystals to renal epithelial cells. We for the first time preliminarily demonstrated that TFDS inhibited MCP-1, OPN and TGF- β expression in this process. Whereas, more in depth experiments are required to further explore the role of MCP-1, OPN and TGF- β involving in the antilithic effects of TFDS.

Conclusion

Taken together, our results indicated that TFDS had beneficial effect on inhibition of CaOx formation in the rat kidney. The mechanism underlying this effect is still ambiguous, but is apparently related to a combination of antioxidant, anti-inflammatory, urine alkalinizing activities, and lowering the concentration of urinary stone-forming constituents. Thus, TFDS might have clinical implications in preventing oxidative renal cell injury and, ultimately, kidney stone formation. The data provide a rationale for the medicinal use of TFDS in nephrolithiasis and identify this agent as a potential source of new antiurolithic drugs.

Acknowledgements We are grateful to Dr. Wenqi Wu (Guangdong Key Laboratory of Urology, Minimally Invasive Surgery Center, the First Affiliated Hospital of Guangzhou Medical University, China) for the kind assistance in this study. This work was supported in part by the Science and Technology Planning Project of Guangdong Province (Grant Nos. 2011B061300077, 2008B030301363), and the Key Project of Traditional Chinese Medicine Bureau of Guangdong Province (Grant No. 20173009), “The Construction of High-level University” Public Projects from Guangzhou University of Chinese Medicine (Grant No. 2017B022).

Compliance with ethical standards

Conflict of interest The authors declare that they have no potential financial competing interests.

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